Archives de pédiatrie 33 (2026) 105583

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Review article

Are children and adolescents with ADHD less physically fit? Findings from a systematic review and meta-analysis

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ARTICLE INFO

Article History:
Received 17 June 2025
Revised 23 March 2026
Accepted 22 May 2026
Available online 25 June 2026

Keywords:
Attention-deficit/hyperactivity disorder
Health-related physical fitness
Cardiorespiratory fitness
Muscular strength
Flexibility
Children
Adolescents

ABSTRACT

Background: Health-related physical fitness is critical from early childhood, yet it is unclear whether children and adolescents with attention-deficit/hyperactivity disorder (ADHD) differ from their typically developing (TD) peers.
Objective: To systematically review and meta-analyze differences in health-related physical fitness between children and adolescents with and without ADHD.
Methods: and Setting: Seven electronic databases were searched for studies comparing cardiorespiratory fitness, muscular strength, and flexibility between children and adolescents with ADHD and TD peers. Methodological quality was evaluated using the Mixed Methods Appraisal Tool.
Results: Thirteen studies including 1,814 participants (560 with ADHD, 1,254 typically developing [TD]) were reviewed. Eleven studies assessed cardiorespiratory fitness. Pooled data from two VO_2peak studies (28 ADHD, 33 TD) showed no significant group differences (mean difference = -4.28 $mL{\cdot}kg^{-1}{\cdot}min^{-1}$; 95% CI: -10.51 to 1.96; p = 0.18; I^2 = 59%). Seven studies using field-based tests revealed mixed outcomes. Pooled data from six studies using the 20-meter Shuttle Run or PACER test (149 ADHD, 949 TD) indicated significantly lower performance in the ADHD group (standardized mean difference = -0.46; 95% CI: -0.78 to -0.14; p = 0.004; I^2 = 47%). Five studies examining heart rate variables reported no significant differences (mean difference = 2.23 bpm; 95% CI: -2.72 to 7.19; p = 0.38; I^2 = 42%). Muscular strength outcomes (six studies) and flexibility (five studies) showed minimal differences between groups (strength: SMD = -0.07, 95% CI: -0.31 to 0.17; flexibility: SMD = -0.16, 95% CI: -0.65 to 0.33), with low-to-moderate heterogeneity.
Conclusion: Children and adolescents with ADHD exhibit comparable levels of muscular strength and flexibility to their TD peers. Evidence on cardiorespiratory fitness is mixed and depends on assessment methods: field-based tests indicate lower performance in ADHD, while laboratory peak oxygen uptake (VO_2peak) and resting heart rate measures show no significant differences.
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Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental condition characterized by problematic levels of inattention, disorganization, and impulsivity, which may occur with varying degrees of hyperactivity [1,2]. Despite its marked heterogeneity and interindividual variability, three common presentation types are typically identified: predominantly inattentive, predominantly hyperactive/impulsive, and combined presentation [1,2].

In the scientific literature, the prevalence of ADHD in children and adolescents ranges from 5.6% to 7.6%, with a higher incidence in males, showing an approximate ratio of 3:1 [3]. Its etiology is complex and multifactorial, resulting from the interaction between genetic variants and environmental factors, with epigenetics potentially playing a key mediating role [4].

Beyond the core clinical features of the disorder, additional health problems frequently emerge. Children and adolescents with ADHD have shown an increased body mass index (BMI), possibly due to fragmentation of the daily rhythm of activity [5]. In addition, later in adulthood, people with ADHD face an increased risk of cardiovascular diseases, sleep problems, and a greater prevalence of physical and mental health problems, which contribute to reduced longevity

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https://doi.org/10.1016/j.arcped.2026.105583
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compared with the general population [6]. In this context, promoting physical activity from an early age is essential, as it is associated with improved life expectancy [7].

The beneficial effects of physical activity are mediated through its impact on dimensions of physical fitness that are directly related to health [8]. Physical fitness refers to the ability to perform physical activity and encompasses a wide range of physiological and psychological attributes, serving as a key indicator of health [9]. According to the model proposed by Caspersen et al [10], physical fitness can be divided into health-related physical fitness and motor-related physical fitness. Health-related physical fitness includes five components: cardiorespiratory fitness, muscular endurance, muscular strength, body composition, and flexibility [10]. These components, particularly muscular strength, muscular endurance, and flexibility, are closely associated with various health outcomes, playing a crucial role in maintaining functional capacity, preventing chronic diseases, and enhancing overall quality of life [11]. Systematic reviews with meta-analyses have shown that greater cardiorespiratory fitness, muscular strength, and flexibility are associated with better health indicators in children and adolescents with typical development (TD) [12,13].

Several studies have examined differences in motor-related physical fitness between children and adolescents with ADHD and their TD peers, consistently reporting poorer performance in those with ADHD [14,15]. In contrast, systematic reviews addressing health-related physical fitness have predominantly focused on body composition [16–19], and to date, no systematic review has specifically examined whether differences exist across other components of health-related physical fitness between children and adolescents with and without ADHD.

This question is relevant because physical fitness is key to health and quality of life, and children with ADHD are often perceived as less fit. Synthesizing existing studies can clarify if this stigma reflects actual differences. For this reason, the aim of this systematic review and meta-analysis was to assess the existence of differences in health-related physical fitness among children and adolescents with and without ADHD.

Methods

This systematic review and meta-analysis adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [20] for transparent and comprehensive reporting. The study protocol was prospectively registered on the Open Science Framework (OSF) at https://doi.org/10.17605/OSF.IO/P7VEA

Search strategy

A comprehensive search was performed in seven electronic databases— Web of Science, PubMed, Scopus, Dialnet Plus, SPORTDiscus, ERIC, and the PEDro Database— encompassing all publications up to March 2025. The search strategy utilized the following Boolean operators and term combinations: ("Physical fitness" OR "fitness levels" OR "motor skills" OR "cardiorespiratory fitness" OR "muscular strength" OR "balance" OR "coordination" OR "endurance" OR "physical activity levels") AND ("Attention-Deficit/Hyperactivity Disorder" OR "ADHD" OR "children with ADHD" OR "adolescents with ADHD").

Eligibility criteria

Studies reporting outcomes related to cardiorespiratory fitness, muscular strength, and flexibility as components of healthy physical fitness in children and adolescents with and without ADHD were considered eligible for inclusion in this review.

Investigations were excluded if: a) the studies did not include children or adolescents with ADHD; b) the research involved participants with multiple pathologies, unless separate data for the ADHD subgroup were provided; c) physical fitness data for ADHD and TD groups were reported but comparative statistical analyses were not conducted; d) baseline data was not informed; and e) the full text of the study was not available.

Study selection

Two researchers independently screened the titles and abstracts of all retrieved studies to assess their eligibility. After completing the review separately, they compared their results and agreed on the final selection. Full texts of studies deemed potentially relevant were then obtained. In cases where eligibility was uncertain, a third reviewer was consulted to resolve discrepancies and reach a final decision. Additionally, Google Scholar was searched for further studies meeting the inclusion criteria. Reference lists of the selected articles and their citing papers were also reviewed to identify any additional relevant studies for inclusion.

Data extraction

For each included study, data were extracted on the authors, year of publication, country, research objective, participant characteristics (sex and age), outcomes assessed, and main findings. One researcher carried out the data extraction, and a second investigator verified its accuracy to ensure consistency and reliability.

Quality appraisal

The methodological quality and risk of bias of the studies included in this review were assessed using the Mixed Methods Appraisal Tool (MMAT) version 2018 [21]. This tool has proven to be useful in systematic reviews that include various study designs [22]. The critical appraisal was conducted independently by two reviewers, with any disagreements resolved through discussion. Item scores may vary as follows: yes (criterion met), no (criterion not met), and unclear (insufficient information provided to assess the criterion).

Statistical analysis

The data extracted from the articles was processed in an Excel spreadsheet, and the meta-analysis was performed using Review Manager (RevMan) software (version 5.4.1). The mean difference or standardized mean difference, along with their 95% confidence intervals (CI), were utilized to estimate effect sizes for quantitative dependent variables. When variability was expressed as standard error, it was converted to standard deviation. In instances where continuous variables were expressed as median and interquartile range, they were converted to mean and standard deviation. The continuous variables were analyzed using a random effects model and the inverse variance method. The confidence interval for the global effect was determined by employing the weighted variance method. The heterogeneity statistic (I2) was utilized to evaluate heterogeneity. An I2 value of 0% indicates the absence of heterogeneity, whilst a higher value indicates greater heterogeneity.

Results

Study selection

A total of 9985 records were initially identified through the database search. After duplicate removal, 7127 titles and abstracts were screened, leading to the retrieval of 335 articles for full-text review. Finally, 10 studies met the inclusion criteria and were incorporated into the systematic review. Additionally, citation tracking of the

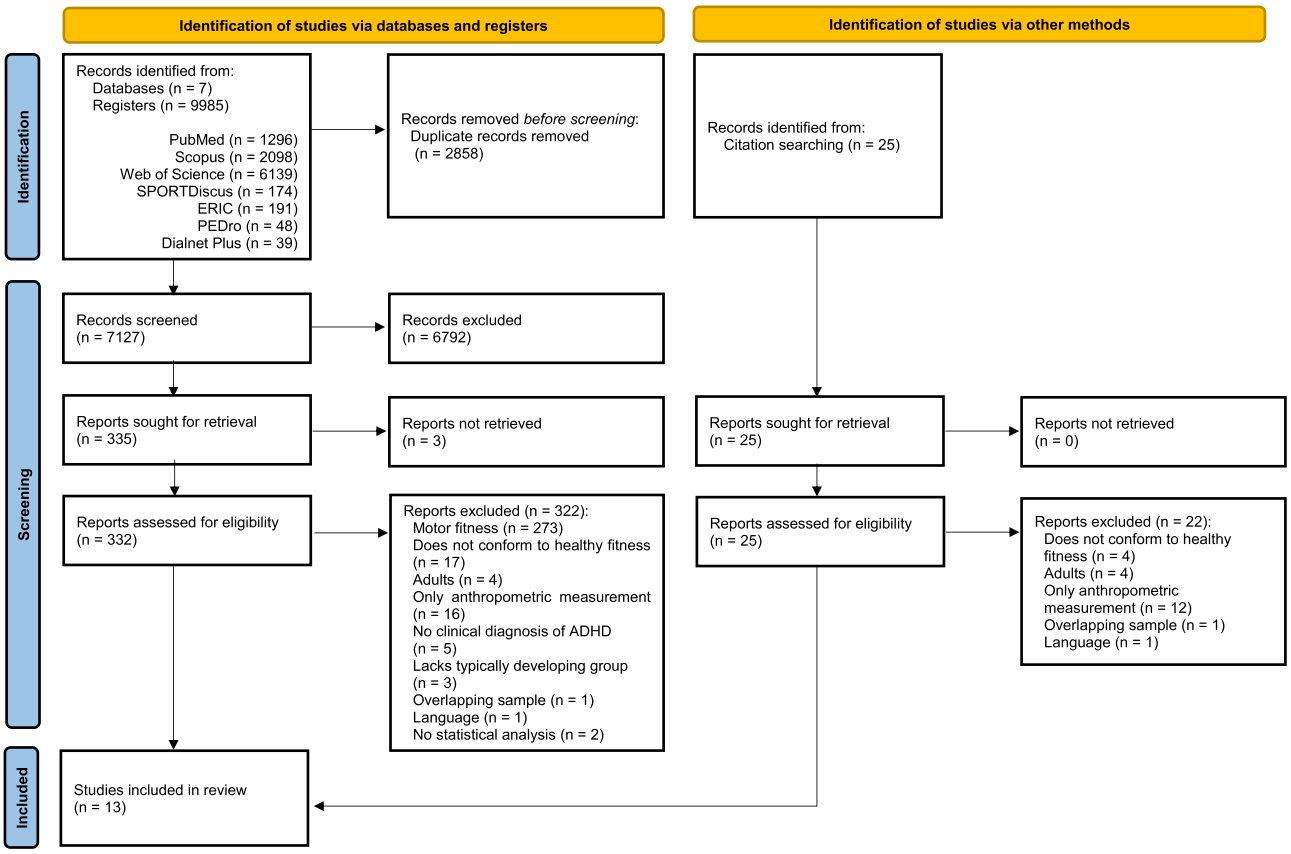


Fig. 1. PRISMA (Preferred reporting items for systematic reviews and meta-analyses) study flow diagram.

selected articles (n = 25) yielded 3 more eligible studies, bringing the final number of included articles to 13 (Fig. 1).

Study characteristics

The reviewed studies were published between 1997 and 2025. Of the total investigations, six primarily aimed to analyze differences in healthy physical fitness parameters between children and adolescents with and without ADHD. The participants were children and adolescents ranging in age from 4–7 years [23] to 11–16 years [24]. Across all included studies, the combined sample size was 1,814 participants (1,153 males and 661 females). Among them, 560 had a clinical diagnosis of ADHD, with 71.61% being male (n = 401). Only four investigations included exclusively male samples [25–28], while the remaining studies involved mixed-gender samples. Sample sizes in individual studies ranged from 18 participants [27] to 892 [29]. Kang et al [30] and Verret et al. [26] further categorized participants with an ADHD diagnosis based on whether they were receiving medication, with a total of 69 participants undergoing pharmacological treatment. Table 1 provides a summary of the main characteristics of the included studies.

Main outcomes

Cardiorespiratory fitness

Eleven studies examined differences in cardiorespiratory fitness parameters between children and adolescents with and without ADHD diagnoses [24–29,31–35].

Peak oxygen uptake (VO_2peak) was assessed by Tantillo et al. [35] and Wigal et al [27] using the Bruce Test and a constant and incremental cycling protocol, respectively. The pooled data from these studies included 28 participants in the ADHD group and 33 in the TD group. The meta-analysis results showed no significant differences between groups (95% CI = -4.28 [-10.51, 1.96]; p = 0.18), with a moderate level of heterogeneity (I^2 = 59%) (Fig. 2).

Maximal oxygen uptake (VO_2max) levels were informed by Buker et al. [31] and Huang et al. [33] using the 20-meter Shuttle Run Test and by Tantillo et al [35] using the Bruce Test. Their findings yielded mixed results. Seven studies assessed performance on specific cardiorespiratory fitness tests [24–26,29,32–34]. No significant differences were found in the Shuttle Run Test (in laps or time), the Bruce Test (time), the Léger Test (time) or in endurance time run [26,32,33]. In the physical working capacity at a heart rate of 170 beats/min (PWC170 test), Ludygova et al [24] reported no significant differences, whereas Ludygova et al [34] found significantly lower performance in the ADHD group (p = 0.002), even after adjusting for BMI (p = 0.002). Martinson et al [29] and Pan et al [25] used the Progressive Aerobic Cardiovascular Endurance Run (PACER Test) and observed significantly lower performance in the ADHD group ($p < 0.001$ and $p < 0.01$, respectively).

The pooled data from six studies that used either the 20-meter Shuttle Run Test (n = 4) or the PACER test (n = 2), including 149 participants with ADHD and 949 TD participants, are presented in Fig. 3. The pooled standardized mean difference was -0.46 (95% CI: -0.78 to -0.14; p = 0.004), indicating a significantly lower cardiorespiratory fitness among children and adolescents with ADHD. Heterogeneity was low to moderate (I^2 = 47%).

Five studies examined heart rate variables, with resting heart rate being the most frequently analyzed [24,28,35]. None of them reported significant intergroup differences. Tantillo et al [35] and Wigal et al [27] also assessed recovery heart rate and peak heart rate, with no significant differences observed. The meta-analysis of the four studies that examined resting heart rate, comparing 76 children and adolescents with ADHD to 88 TD peers, showed no significant

Table 1
Descriptive characteristics of the included studies.

First Author (Year), Country	Objective	Sample	Outcomes	Results
Buker et al. [31] Country: Turkey	To investigate both the physical fitness levels and the dynamic balance abilities of medication-naïve Turkish children with ADHD compared to their peers without ADHD.	Participants (n): 43 (24ADHD; 19TD) Sex (n): 35M; 8F ADHD (20M; 4F) TD (15M; 4F) Age, years (range; mean; SD): 7-12; 8.88 ± 1.59 years ADHD: 9.00 ± 1.69 years TD: 8.68 ± 1.57 years	Cardiorespiratory fitness: 20-meter Shuttle Run (ml/kg/min) Strength: Standing Broad Jump (cm) Hand Grip (kg) Sit-up (rep) Bent Arm Hang (s) Flexibility: Sit and Reach (cm)	Inter-group (p< 0.05): 20m Shuttle Run: ns (p= 0.318) ADHD: (42.95; IQR [41.50/45.70]) vs TD: (45.45; IQR [43.78/109.45]) Standing broad jump: ns (p= 0.525) ADHD: (91.00; IQR [78.50/104.75]) vs TD: (92.00; IQR [82.00/103.00]) Hand grip: ns (p= 0.138) ADHD: (13.50; IQR [10.25/17.50]) vs TD: (15.50; IQR [14.00/18.00]) Sit-up: ns (p= 0.649) ADHD: (11.00; IQR [2.25/13.75]) vs TD: (11.00; IQR [6.00/15.00]) Bent arm hang: ns (p= 0.480) ADHD: (0; IQR [0/25.18]) vs TD: (0; IQR (0/41.60]) Sit and Reach ns (p= 0.835) ADHD: (14.00; IQR [(8.25/16.00)] vs TD: (14.00; IQR [(7.00/19.00)]
Colombo-Dougovito [32] Country: USA	To analyze the possible differences of the physical fitness performance of elementary-aged students with and without ADHD.	Participants (n): 51 (8ADHD; 43TD) Sex (n): 27M; 24F ADHD (5M; 3F) TD (22M; 21F) Age, years (mean): ADHD: 7.75 years TD: 8.23 years	Cardiorespiratory fitness: 20-meter Shuttle Run (s) Endurance Run (s) Strength: Curl-up (rep) Push-up (rep) Flexed-arm hang (s) Pull-up (rep) Flexibility: Sit and Reach (cm)	Inter-group (p< 0.05): 20m Shuttle run: ns (p= 0.890) ADHD: (12.56 ± 1.23) vs TD: (13.62 ± 1.26) Endurance run: ns (p= 0.579) ADHD: (235.67 ± 107.96) vs TD: (261.02 ± 126.57) Curl-Up: ns (p= 0.207) ADHD: (30.56 ± 10.20) vs TD: (27.00 ± 6.95) Push-Up: (p= 0.029) favouring ADHD ADHD: (34.48 ± 11.49) vs TD: (10.95 ± 1.69) Flexed-arm hang: ns (p= 0.113) (10.06 ± 6.36) vs TD: (7.43 ± 3.93) Pull-Up: (p= 0.012) favouring ADHD ADHD: (3.33 ± 4.64) vs TD: (1.07 ± 1.58) Sit and Reach: ns (p= 0.303) ADHD: (30.0 ± 4.38 vs TD: (28.11 ± 5.05)
Curzon et al. [23] Country: USA	To assess differences in child physical health outcomes and metrices associated with obesity in a sample of predominantly Hispanic/Latinx young children with and without ADHD.	Participants (n): 223 (127ADHD; 96TD) Sex (n): 147M; 76F Age, years (range; mean; SD): 4-7; 5.48 ± 0.74 years Cohort 1: Participants (n): 55 (33ADHD; 22TD) Sex (n): ADHD (24M; 9F) TD (18M; 4F) Age, years (mean; SD): ADHD: 5.63 ± 0.87 years TD: 5.23 ± 0.86 years Cohort 2: Participants (n): 168 (94ADHD; 74TD) Sex (n): ADHD (68M; 26F) TD (37M; 37F) Age, years (mean; SD): ADHD: 5.52 ± 0.61 years TD: 5.43 ± 0.79 years	Strength: Side jumps (rep)	Inter-group (p< 0.05): Cohort 1: Side jumps: ns (p> 0.05) ADHD: (14.94 ± 6.85) vs TD: (17.92 ± 1.12) Cohort 2: Side jumps: ns (p> 0.05) ADHD: (15.90 ± 7.57) vs TD: (18.32 ± 9.29)

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Table 1 (*Continued*)

First Author (Year), Country	Objective	Sample	Outcomes	Results
Huang et al. [33] Country: China	To investigate the effects of rope skipping exercise on the working memory and cardiorespiratory fitness of children with ADHD.	Participants (n): 57 (38ADHD; 19TD) [22ADHDRope Exercise; 16ADHDSport Game] Sex (n): 38M; 19F ADHD (27M; 11F) TD (11M; 8F) Age, years (range; mean; SD): 6-12 ADHDRope Exercise: 10.18 ± 1.10 years ADHDSport Game: 8.94 ± 0.56 years TD: 8.94 ± 0.56 years	Cardiorespiratory fitness: 20-meter Shuttle Run Test (laps) VO2max by Shuttle Run (ml/kg/min)	Inter-group (p< 0.05): 20m Shuttle run: ns (p= 0.051) ADHDRope Exercise: (14.64 ± 5.61) vs TD: (20.81 ± 9.40) ns (p= 0.051) ADHDSport Game: (14.56 ± 4.10) vs TD: (20.81 ± 9.40) VO2max: (p< 0.001) not favouring ADHDRope Exercise ADHDRope Exercise: (42.28 ± 2.27) vs TD: (45.80 ± 2.71) (p= 0.017) not favouring ADHDSport Game ADHDSport Game: (43.70 ± 2.39) vs TD: (45.80 ± 2.71) ns (p= 0.083) ADHDRope Exercise: (42.28 ± 2.27) vs ADHDSport Game (43.70 ± 2.39)
Iaboni et al. [28] Country: Canada	To analyze the patterns of psychophysiological response to reward and extinction.	Participants (n): 36 (18ADHD; 18TD) Sex (n): 36M Age, years (range; mean; SD): 8-13 ADHD: 127.1 ± 17.25 months TD: 132.1 ± 18.97 months	Cardiorespiratory fitness: Resting heart rate (bpm)	Inter-group (p < 0.05): Resting heart rate: ns (p> 0.05) ADHD: (84.9 ± 2.68) vs TD: (78.6 ± 2.92)
Kang et al. [30] Country: Korea	To investigate the effect of methylphenidate on clinical symptoms, growth, and physical fitness in Korean children.	Participants (n): 179 (119ADHD; 60TD) [ADHDMED:50; ADHDN-MED: 69] Sex (n): 144M; 35F ADHDMED (43M; 7F) ADHDN-MED (56M; 13F) TD (45M; 15F) Age, years (range; mean; SD): 7-12 ADHDMED: 9.7 ± 1.6 years ADHDN-MED: 9.5 ± 1.5 years TD: 9.2 ± 1.9 years	Strength: Back strength (kg) Sit-up (rep) Flexibility: Standing trunk flexion test (cm)	Inter-group (p< 0.05): Back strength: ns (p= 0.83) ADHDMED: (21.4 ± 13.5) vs ADHDN-MED: (19.5 ± 9.5) vs TD: (22.7 ± 10.7) Sit up: ns (p= 0.74) ADHDMED: (30.3 ± 19.1) vs ADHDN-MED: (32.4 ± 15.8) vs TD: (33.6 ± 15.2) Standing trunk flexion: ns (p= 0.83) ADHDMED: (4.2 ± 8.3) vs ADHDN-MED: (4.8 ± 7.9) vs TD: (6.8 ± 6.0)
Ludyga et al. [24] Country: Switzerland	To investigate cognitive flexibility and task-related heart rate variability following moderately intense aerobic exercise and after watching a video in both children with ADHD and TD controls.	Participants (n): 34 (16ADHD; 18TD) Sex (n): 22M; 12F ADHD (11M; 5F) TD (11M; 7F) Age, years (range; mean; SD): 11-16 ADHD: 12.8 ± 1.8 years TD: 13.5 ± 1.3 years	Cardiorespiratory fitness: PWC170 (W/kg) Resting heart rate (bpm)	Inter-group (p< 0.05): PWC170: ns (p= 0.557) ADHD: (2.4 ± 0.7) vs TD: (2.5 ± 0.5) Resting heart rate: ns (p= 0.116) ADHD: (75.0 ± 9.6) vs TD: (69.2 ± 11.1)
Ludyga et al. [34] Country: Switzerland German	To compare physical activity and sports participation between children with ADHD and TD controls.	Participants (n): 120 (60ADHD; 60TD) Sex (n): 81M; 39F ADHD (42M; 18F) TD (39M; 21F) Age, years (mean; SD): ADHD: 10.5 ± 1.2 years TD: 11.1 ± 1.4 years	Cardiorespiratory fitness: PWC170 (W/kg) PWC170 adjusted for BMI (W/kg)	Inter-group (p< 0.05): PWC170: (p= 0.002) not favouring ADHD ADHD: (NR) vs TD: (NR) PWC170 adjusted for BMI: (p= 0.002) not favouring ADHD ADHD: (NR) vs TD: (NR)
Martinson et al. [29] Country: USA	To examine the performance on the PACER test by children with and without ADHD, and also investigate the possible influences of age, sex, BMI and physical activity.	Participants (n): 892 (55ADHD; 837TD) Sex (n): 466M; 426F Age, years (range; mean; SD): 11-14; 12.25 ± 0.94 years	Cardiorespiratory fitness: PACER test (laps)	Inter-group (p< 0.05): PACER test: (p< 0.001) not favouring ADHD ADHD: (23.70 ± 12.39) vs TD: (29.04 ± 16.60)

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Table 1 (*Continued*)

First Author (Year), Country	Objective	Sample	Outcomes	Results
Pan et al. [25] Country: Taiwan	This study explored how a 12-week simulated developmental horse-riding program combined with fitness training influenced the motor proficiency and physical fitness of children with ADHD.	Participants (n): 48 (24ADHD; 24TD) [ADHDTRAIN:12; ADHDNO TRAIN: 12] Sex (n): 48M Age, years (range; mean; SD): 7-14 ADHDTRAIN: 9.63 ± 2.48 years ADHDNO TRAIN: 9.38 ± 2.69 years TD: 9.57 ± 2.50 years	Cardiorespiratory fitness: PACER test (laps) Strength: Push-up isometric (s) Curl-up (rep) Flexibility: Back-saver Sit and Reach – right leg (cm) Back-saver Sit and Reach – left leg (cm)	Inter-group (p< 0.01): PACER test: (p< 0.01) not favouring ADHDTRAIN ADHDTRAIN: (12.582 ± 6.39) vs TD: (22.71 ± 14.94) (p< 0.01) not favouring ADHDNO TRAIN ADHDNO TRAIN: (11.25 ± 5.34) vs TD: (22.71 ± 14.94) Push up: ns (p> 0.01) ADHDTRAIN: (33.25 ± 12.26) vs TD: (39.25 ± 2.69) ns (p> 0.01) ADHDNO TRAIN: (34.42 ± 12.03) vs TD: (39.25 ± 2.69) Curl up: ns (p> 0.01) ADHDTRAIN: (3.33 ± 6.96) vs TD: (6.04 ± 8.53) ns (p> 0.01) ADHDNO TRAIN: (4.58 ± 7.91) vs TD: (6.04 ± 8.53) Back-saver sit and reach – Right: ns (p> 0.01) ADHDTRAIN: (27.04 ± 6.91) vs TD: (25.23 ± 4.98) ns (p> 0.01) ADHDNO TRAIN: (26.71 ± 7.45) vs TD: (25.23 ± 4.98) Back-saver sit and reach – Left: ns (p> 0.01) ADHDTRAIN: (26.50 ± 8.22) vs TD: (24.48 ± 6.12) ns (p> 0.01) ADHDNO TRAIN: (26.58 ± 7.53) vs TD: (24.48 ± 6.12)
Tantillo et al. [35] Country: USA	The effects of exercise in ADHD children on spontaneous blink rate, acoustic startle eye blink response, and motor impersistence.	Participants (n): 43 (18ADHD; 25TD) Sex (n): 21M; 22F ADHD (10M; 8F) TD (11M; 14F) Age, years (range; mean; SD): 8-12 ADHDM: 9.80 ± 1.55 years ADHDF: 10.24 ± 1.77 years TDM: 10.27 ± 1.56 years TDF: 9.71 ± 1.55 years	Cardiorespiratory fitness: Bruce test: Resting Heart rate (bpm) VO2peak (ml/kg/min) Recovery Heart rate (bpm) VO2max (ml/kg/min)	Inter-group (p< 0.05): Resting heart rate: Males: ns (p> 0.05) ADHD: (80.02 ± 10.70) vs TD: (87.58 ± 12.74) Females: ns (p> 0.05) ADHD: (76.89 ± 14.54) vs TD: (80.02 ± 9.92) Bruce VO2peak: Males: ns (p> 0.05) ADHD: (41.10 ± 7.46) vs TD: (40.61 ± 10.64) Females: ns (p> 0.05) ADHD: (31.85 ± 5.23) vs TD: (43.32 ± 13.54) Bruce Recovery heart rate: Males: ns (p> 0.05) ADHD: (97.03 ± 13.34) vs TD: (85.45 ± 9.75) Females: ns (p> 0.05) ADHD: (92.33 ± 11.55) vs TD: (101.3 ± 7.00) Bruce VO2max: Males: ns (p> 0.05) ADHD: (31.32 ± 3.97) vs TD: (27.67 ± 6.73) Females: ns (p> 0.05) ADHD: (20.24 ± 3.51) vs TD: (24.65 ± 4.43)

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Table 1 (*Continued*)

First Author (Year), Country	Objective	Sample	Outcomes	Results
Verret et al. [26] Country: Canada	To assess fitness and gross motor performance of children with ADHD, including users and nonusers of methylphenidate medication.	Participants (n): 70 (43ADHD; 27TD) [ADHDMED:19; ADHDN-MED: 24] Sex (n): 70M Age, years (range; mean; SD): 7-12 ADHDMED: 10.0 ± 1.1 years ADHDN-MED: 9.4 ± 1.5 years TD: 9.7 ± 1.3 years	Cardiorespiratory fitness: Bruce test (min) Léger test (stage) Resting heart rate (bpm) Strength: Push-up (rep) Sit-up (rep) Flexibility: Sit and Reach (cm)	Inter-group (p< 0.05): Bruce test: ns (p= 0.200) ADHDMED: (10.7 ± 2.7) vs ADHDN-MED: (10.5 ± 2.5) vs TD: (12.4 ± 3.4) Léger test: ns (p= 0.327) ADHDMED: (2.9 ± 1.8) vs ADHDN-MED: (2.0 ± 0.9) vs TD: (3.3 ± 1.4) Resting heart rate: ns (p= 0.465) ADHDMED: (85.4 ± 14.7) vs ADHDN-MED: (84.8 ± 13.8) vs TD: (80.8 ± 12.0) Push-up: ns (p= 0.125) ADHDMED: (8.8 ± 6.4) vs ADHDN-MED: (11.7 ± 8.8) vs TD: (13.0 ± 5.9) Sit-up: ns (p= 0.125) ADHDMED: (32.1 ± 14.5) vs ADHDN-MED: (29.8 ± 14.7) vs TD: (33.2 ± 7.5) Sit and Reach: ns (p= 0.113) ADHDMED: (22.1 ± 7.2) vs ADHDN-MED: (23.9 ± 7.37) vs TD: (25.9 ± 4.5)
Wigal et al. [27] Country: USA	To examine differences in catecholamine response to exercise between children who had received a diagnosis of ADHD and age- and gender-matched controls.	Participants (n): 18 (10ADHD; 8TD) Sex (n): 18M Age, years (range; mean; SD): ADHD: 7-10; 8.4 ± 0.4 years TD: 7-11; 8.6 ± 0.5 years	Cardiorespiratory fitness: Ramp-type and constant work on cycle ergometer: Peak VO2A (l/min) Peak VO2B (ml/kg/min) Peak work rate (watts) Lactate threshold (mg/dl) Peak lactate (mg/dl) Peak heart rate (bpm)	Inter-group (p< 0.05): Peak VO2A: ns (p> 0.05) ADHD: (1.01± 0.07) vs TD: (1.11 ± 0.09) Peak VO2B: ns (p> 0.05) ADHD: (30.2 ± 1.7) vs TD: (32.8 ± 2.0) Peak work rate: ns (p> 0.05) ADHD: (85.7 ± 5.7) vs TD: (98.7 ± 7.1) Lactate threshold: ns (p> 0.05) ADHD: (0.78 ± 0.06) vs TD: (0.79 ± 0.05) Peak lactate: ADHD < TD (p< 0.005) ADHD: (NR) vs TD: (NR) Peak heart rate: ns (p> 0.05) ADHD: (178.1 ± 5.1) vs TD: (189.4 ± 3.1)

Abbreviations: ADHD: Attention Deficit/Hyperactivity Disorder; BMI: body mass index; bpm: beats per minute; cm: centimeters; F: female; IQR: interquartile range; kg: kilograms; l/min: liters/per minute; M: male; MED: medication; min: minutes; ml/kg/min: milliliters per kilogram per minute; N-MED: not medication; NR: not reported; ns: not significant; Rep: repetitions; s: seconds; SD: standard deviation; TD: Typically Developing; W/kg: watts per kilogram.
Note: results are presented as mean ± SD or median (IQR), as reported in the original studies.

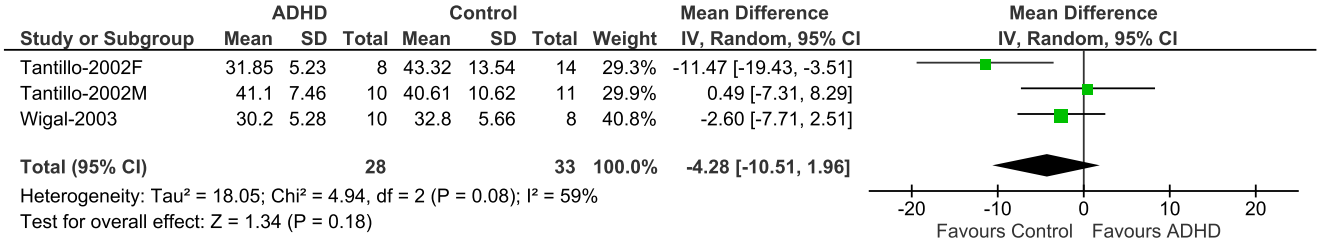


Fig. 2. Forest plot of the comparison of VO_2peak differences between children and adolescents with ADHD and those with TD.
ADHD: Attention-deficit/hyperactivity disorder; VO2peak: Peak oxygen uptake; TD: Typically developing; CI: confidence interval; SD: standard deviation.

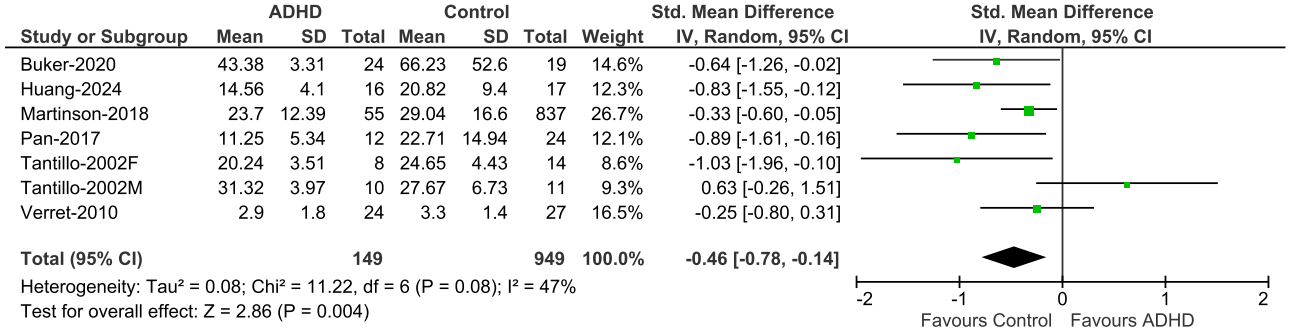


Fig. 3. Forest plot of the comparison of cardiorespiratory fitness performance in the 20m shuttle run test and PACER between children and adolescents with ADHD and those with TD.
ADHD: Attention-deficit/hyperactivity disorder; TD: Typically developing; PACER: Progressive aerobic cardiovascular endurance run; CI: confidence interval; SD: standard deviation.

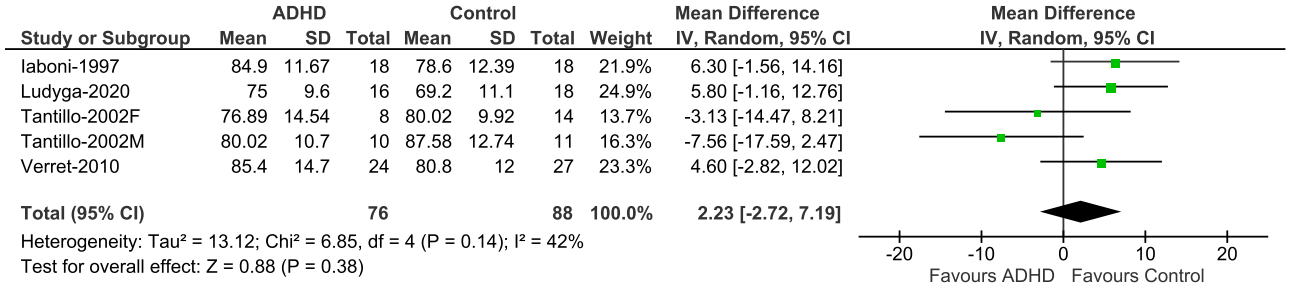


Fig. 4. Forest plot of the comparison of resting heart rate differences between children and adolescents with ADHD and those with TD.
ADHD: Attention-deficit/hyperactivity disorder; TD: Typically developing; CI: confidence interval; SD: standard deviation.

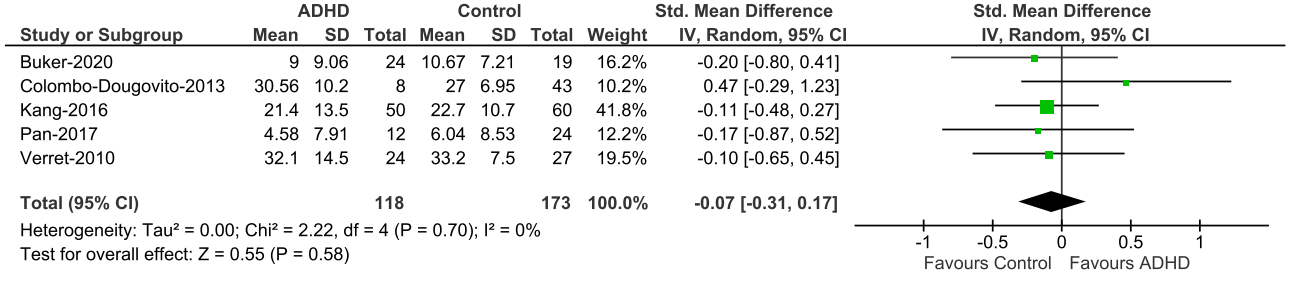


Fig. 5. Forest plot of the comparison of sit-up and curl-up between children and adolescents s with ADHD and those with TD.
ADHD: attention-deficit/hyperactivity disorder; TD: Typically developing; CI: confidence interval; SD: standard deviation.

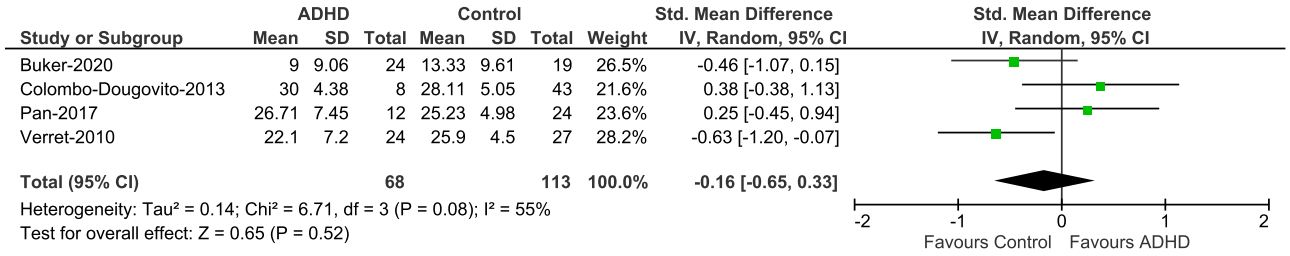


Fig. 6. Forest plot of the comparison of flexibility differences between children and adolescents with ADHD and those with TD.
ADHD: Attention-deficit/hyperactivity disorder; TD: typically developing; CI: confidence interval; SD: standard deviation.

differences, mean difference 2.23 bpm (95% CI -2.72 to 7.19) (p = 0.38), with a moderate to high level of heterogeneity (I^2 = 42%) (Fig. 4).

Muscular strength

Six studies compared muscular strength parameters between children and adolescents with and without ADHD [23,25,26,30–32].

Muscular endurance was the most frequently assessed strength-related physical fitness component. The tests used included the sit-up [26,30,34], curl-up [25,32], push-up [26,32], pull-up [32], bent-arm hang [31], flexed arm hang [32], and side jumps [23]. Only Colombo-Dougovito et al [32] reported significant differences in favor of the ADHD group in the push-up (p = 0.029) and pull-up (p = 0.012) tests. The meta-analysis results concerning the curl-up and sit-up revealed a standardized mean difference of -0.07 (CI95% -0.31 to 0.17: p=0.58: I2=0%) (Fig. 5).

Three studies administered isometric strength tests, including the hand grip strength, back strength test, and isometric push-up [25,30,31], and no significant differences between participants with and without ADHD were reported. Similar findings were reported by Buker et al [31] following an assessment of lower-body explosive strength using the Standing Broad Jump test.

Flexibility

Flexibility differences were assessed in five studies [25,26,30–32]. For comparative analysis, four investigations used the Sit & Reach test [25,26,31,32], while Kang et al [30] employed the Standing Trunk Flexion test. None of the investigations reported the existence of significant differences. This trend was confirmed by a meta-analysis combining the four studies that assessed flexibility through the Sit & Reach test, for a standardized mean difference of -0.16 (CI 95% -0.65 to 0.33; p= 0.52; I^2 = 55%) (Fig. 6).

A sensitivity analysis with a different effect size estimate (Hedges' g) and prediction intervals did not reveal major changes in our results (Supplementary material. Table S1).

Methodological quality

As shown in Table 2, the quality was highly heterogeneous across studies. Five studies were quantitative non-randomized [23,25,29,32,33]. In general, none of these studies included a sample representative of the target population. In contrast, all of them used appropriate measurements for both the outcome and the intervention.

In the case of the quantitative descriptive studies, the most frequent source of bias was related to the relevance of the sampling strategy to the research question, with only one study meeting this criterion [24]. In addition, only those investigations conducted by Buker et al [31], Kang et al [30], Ludygova et al [24], and Ludygova et al [34] included samples representative of the target population. Conversely, all studies employed appropriate tools to assess outcomes and interventions, and the interventions were generally delivered as planned.

Discussion

This systematic review and meta-analysis aimed to assess the existence of differences in health-related physical fitness between children and adolescents with and without ADHD. The obtained findings provide insightful information on healthy fitness in this population, which may guide future interventions and assessment strategies for children and adolescents with ADHD.

Cardiorespiratory fitness is a key health indicator in children and adolescents, as low levels have been significantly associated with a higher risk of developing metabolic syndrome [36], as well as other chronic diseases and increased mortality in adulthood [37]. This component of health-related physical fitness was the most frequently analyzed outcome, addressed in eleven of the included investigations, although findings were inconsistent. Studies using field-based tests to estimate VO_2max—such as the PACER and Shuttle Run tests, which are widely used for children and adolescents—generally indicated lower cardiorespiratory fitness in ADHD groups compared with TD peers. In contrast, studies that assessed VO_2peak in laboratory settings did not show consistent differences between groups. This distinction is important, as laboratory tests are considered the gold standard for assessing cardiorespiratory fitness, and VO_2peak is recognized as the most reliable indicator in children [38]. However, evidence from laboratory assessments remains limited, as the meta-analysis included only two investigations. Interestingly, resting heart rate—another validated biomarker of cardiorespiratory fitness [39]—did not differ significantly between groups. In addition, heterogeneity across studies may have played a role in the variability of these findings, as factors such as ADHD subtype, medication status, or sex could have acted as potential moderators. Furthermore, although the included studies generally employed appropriate statistical analyses and assessment methods, their samples presented notable limitations. Specifically, as most studies were not primarily designed to compare cardiorespiratory fitness between participants with and without ADHD, their sampling strategies were not always adequate or representative, thereby restricting the generalizability of the findings. Consequently, the evidence regarding differences in cardiorespiratory fitness between children and adolescents with ADHD and their TD peers remains inconclusive.

Muscular fitness during childhood and adolescence is a key health indicator and a protective factor against the development of diseases in adulthood [12,40]. In children with ADHD, greater muscular strength has also been associated with improved cognitive performance and reduced hyperactive-impulsive behaviors in the school setting [41]. In this review, muscular endurance was the most frequently assessed component, with findings from the meta-analysis indicating that no significant differences were observed between the analyzed groups. This result may be due to the fact that these tests

Table 2
MMAT quality appraisal results.

First author, year	3. Quantitive non-randomized					4. Quantitative descriptive				
	3.1. Are the participants representative of the target population?	3.2. Are measurements appropriate regarding both the outcome and intervention (or exposure)?	3.3. Are there complete outcome data?	3.4. Are the confounders accounted for in the design and analysis?	3.5. During the study period, is the intervention administered (or exposure occurred) as intended?	4.1. Is the sampling strategy relevant to address the research question?	4.2. Is the sample representative of the target population?	4.3. Are the measurements appropriate?	4.4. Is the risk of nonresponse bias low?	4.5. Is the statistical analysis appropriate to answer the research question?
Buker et al. [31]						N	Y	Y	Y	Y
Colombo-Dougovito [32]	N	Y	Y	Y	Y					
Curzon et al. [23]	N	Y	N	Y	Y					
Huang et al. [33]	N	Y	Y	N	Y					
Iaboni et al. [28]						N	C	Y	C	Y
Kang et al. [30]						C	Y	Y	C	Y
Ludyga et al. [24]						Y	Y	Y	Y	Y
Ludyga et al. [34]						N	Y	Y	Y	Y
Martinson et al. [29]	N	Y	Y	N	Y					
Pan et al. [25]	N	Y	Y	Y	Y					
Tantillo et al. [35]						C	C	Y	Y	Y
Verret et al. [26]						N	N	Y	Y	Y
Wigal et al. [27]						N	N	Y	N	Y

Y = YES, N = NO, C = Cannot tell. Mixed Methods Appraisal Tool (MMAT), version 2018 (Hong et al., 2018)

involve repetitive movements of short duration and moderate-to-high intensity in a closed skill context, which may align better with the behavioral profile of children with ADHD, who tend to prefer brief and dynamic tasks over open skills in changing environments [42,43]. In this regard, studies assessing isometric and explosive strength reported no significant differences, indicating that children and adolescents with ADHD generally exhibit comparable strength levels to their TD peers.

However, given the methodological weaknesses identified through the MMAT assessment, these null findings should be interpreted with caution, as limitations in sample representativeness and sampling adequacy may have reduced the studies' ability to detect subtle between-group differences.

Flexibility provides essential health benefits, including increased range of motion, injury prevention and reduction, pain relief, and improved coordination—factors that are fundamental throughout the lifespan [44]. In this review, flexibility was the least studied component of physical fitness. The meta-analysis found no significant differences in flexibility between children and adolescents with ADHD and their TD counterparts. These findings suggest that ADHD does not significantly affect flexibility, at least during childhood and adolescence. This may be due to the fact that flexibility largely depends on structural and anatomical factors [45], which are unlikely to be directly influenced by the core symptoms of ADHD. However, given that only two studies were specifically designed to examine between-group differences, these findings should be interpreted with caution, as sampling limitations likely undermined their robustness despite strengths in assessment, statistical analyses, and intervention implementation.

This seems to be the first systematic review and meta-analysis to investigate differences in health-related physical fitness between children and adolescents with and without ADHD. While the findings do not provide conclusive evidence regarding differences in cardiorespiratory fitness, they suggest that strength and flexibility levels are largely comparable between the two groups. It should be noted, however, that most included studies involved prepubertal or early pubertal participants, a developmental stage in which differences in physical fitness components tend to be less distinguishable compared with the pubertal years [46].

Despite the novelty of this review, several limitations should be acknowledged. First, only six of the included studies explicitly aimed to assess differences in health-related physical fitness between children and adolescents with and without ADHD. Second, although most studies used valid instruments and procedures, many were constrained by small sample sizes, uneven sex distribution, or poorly defined inclusion criteria, particularly concerning ADHD severity and medication status. Additionally, some of the studies presented significant risks of bias, which may have influenced the reliability of the findings. Third, although meta-analyses were performed, they were based on a small number of studies and exhibited heterogeneity, which precluded a formal assessment of publication bias. Fourth, body composition was not included, despite being a core component of health-related physical fitness, and its omission may have influenced the other outcomes examined. Lastly, the exclusion of grey literature may have narrowed the scope of the review, potentially omitting relevant data that could have strengthened the overall findings.

Conclusion

The findings of this review suggest that children and adolescents with ADHD exhibit comparable levels of muscular strength and flexibility to their TD peers. However, evidence on cardiorespiratory fitness is mixed: field-based tests suggest lower performance in ADHD, whereas laboratory-based VO_2peak measurements and resting heart rate show no significant differences. These discrepancies highlight

the influence of assessment methods and indicate that current evidence is insufficient to draw firm conclusions about cardiorespiratory fitness in this population. Further high-quality research, particularly studies incorporating VO_2peak assessments, is needed to determine whether meaningful disparities in cardiorespiratory fitness exist between these populations.

Funding statement

This research did not receive external funding.

Declaration of competing interest

The authors report there are no competing interests to declare.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.arcped.2026.105583.

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